

ENGR 415 SCRT Report

Structures/Characterization of Engineering Filaments for Aerospace Components

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Figure 1: Specimen in Instron Machine Tensile Test

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0. Abstract/Design Specification

This project focused on the material characterization of additively manufactured polymer filament for use in structural rocket components. The overarching team goal is to compete in the International Rocketry Competition by developing a high-performance, reliable launch vehicle. Within this framework, the manufacturing sub-team's objective is to establish repeatable fabrication processes that produce structurally sound and optimized components capable of integrating and retaining all major rocket subsystems. The specific subsystem goal of this work was to experimentally characterize the mechanical properties of printed materials in order to define realistic design allowables for structural components manufactured in-house.

The team's design philosophy emphasizes data-driven engineering, conservative structural assumptions, and iterative validation. Rather than relying solely on manufacturer datasheets, this project sought to quantify the actual mechanical properties achievable through our additive manufacturing processes. This approach ensures that structural analyses reflect real-world manufacturing variability and anisotropic behavior inherent to fused filament fabrication.

The primary system design feature of this work was the development of a standardized testing framework for additively manufactured specimens. This included defining baseline specimen geometries, identifying critical material properties for structural design, and establishing repeatable test procedures. Tensile testing was performed to determine elastic modulus, ultimate tensile strength, and failure strain. Dimensional measurements were conducted to evaluate manufacturing consistency and specimen quality. These properties form the foundation for structural sizing, safety factor selection, and optimization of rocket components.

The most significant result of this study was the consistent observation that experimentally measured mechanical properties were lower than the values reported on manufacturer datasheets, as seen in table 1. This discrepancy highlights the influence of print orientation, layer adhesion, infill parameters, and process variability on final part performance. Compared to baseline assumptions that used published material properties, the experimentally derived values provide a more conservative and realistic basis for structural analysis. Agreement between repeated tensile tests demonstrated acceptable consistency within our controlled process, validating the reliability of the developed testing methodology.

Major challenges included controlling print variability and ensuring proper calibration of testing equipment to minimize experimental error. These were addressed through machine calibration, multiple specimen trials, and repeatability checks. The resulting dataset establishes credible design allowables specific to our team's manufacturing capabilities.

System performance is defined by the accuracy and repeatability of the material data collected. Test results showed internal consistency and predictable failure modes, though at reduced strength relative to nominal material specifications. These findings directly inform structural redesign and safety factor refinement across the rocket.

Looking forward, future work should expand characterization to include compressive, shear, fatigue, and temperature-dependent properties, as well as investigation of print parameter optimization to close the gap between datasheet and achieved performance. Building upon this foundation will enable increasingly optimized, lightweight, and reliable structural components for future competition vehicles.

Table 1: Comparing Data Sheet Values to Average Collected Values

Source	Average Tensile Strength	Average Modulus of Elasticity
Data Sheet	27 +/- 5 MPa	1950 +/- 210 MPa
Collected Values	45.91 MPa	1629.32 MPA

0.1 Specification Sheet

In this section are the tools and materials used to complete the process and goals of this capstone project. Table 2 and table 3 list the machinery and equipment that supported both the manufacturing and the testing process, as well as the filament tested.

Table 2: Tools Needed to Complete the Capstone

Tool	Role
P1S 3D Printer	Create test specimen
Crealty Space Pi Filament Dryer	Dry out filament for printing
Caliper	Dimensional measurement
Annealing Oven	Anneal test specimen
Instron Tensile Tester	Perform tensile test

Table 3: Material

Material	Link to Filament
Bambu PA6-GF*	https://us.store.bambulab.com/products/pa6-gf

*For this project, PA6-GF was tested as the currently available material.

0.2 Manufacturing

Below in table 4 is the manufacturing process for the test specimen used in this capstone project.

Table 4: Part Manufacturing

Part	Materials	Color	Process	Time to Produce
Test Specimen	PA6-GF	Orange	3D Printing	3 Hours

0.3 Summary of Results

Below in table 5 and table 6 are the baseline and the results of our testing. As seen in the two tables, the material performed better in the tensile strength test than what was listed in the data sheet, while the young's modulus performed just under the expected values.

Table 5: Baseline data sheet results

Parameter	Values
Young's Modulus (Z)	1950 +/- 210 MPa
Tensile Strength (Z)	27 +/- 5 MPa
Shrink Factors	Unknown

Table 6: Summary Table of Testing Results

Annealing Time (hrs)	Shrink Factors	Tensile Strength (MPa)	Young's Modulus (MPa)
0	0.9929	48.75	1655.15
3	0.9926	47.12	1619.51
6	0.9922	48.01	1551.23
9	0.9917	43.33	1608.63
12	0.9913	42.37	1712.06

1. Project Description

1.1 Introduction

This project evaluates the application of additive manufacturing for producing structural components, including couplers and nosecone interfaces, in the OSU AIAA IREC rocket, see Fig. 2. The objective is to determine whether 3D-printed materials can reduce manufacturing time, improve dimensional accuracy, and support rapid design iteration compared to the current carbon fiber layup process. Testing will assess the strength, precision, and reliability of additively manufactured parts to verify their suitability as replacements for conventionally fabricated components within the rocket structure.



Figure 2: SCRT Rocket Concept

1.2 Requirements

The objective of this project is to design, manufacture, and operate a fully student-researched and developed rocket propulsion system that meets both Team SCRT performance objectives and the Intercollegiate Rocket Engineering Competition (IREC) requirements, with the ultimate goal of achieving top placement in the competition. The overall system must:

- Carry a minimum payload of 4.4 lb (2.0 kg) to a target apogee of 10,000 ft (3,048 m).
- Be entirely designed, analyzed, and fabricated by Team SCRT, with no commercial-off-the-shelf (COTS) propulsion hardware.
- Meet all dimensional, mass, and safety constraints defined by the IREC 2025 rules and guidelines.

1.2.1 Team Requirements

To ensure successful project execution, all team members are expected to participate actively in the design, manufacturing, and testing phases. Team SCRT maintains organization and coordination through

weekly general meetings, as well as dedicated subteam meetings. Collaborative workflows are enabled via a Discord server for real-time communication, shared documentation on Google Drive, and a cloud-based CAD system (Onshape), allowing concurrent design collaboration and robust configuration management.

1.2.2 Subteam Requirements

The Structures Subteam is responsible for the rocket's primary physical framework, ensuring that the vehicle maintains structural integrity under aerodynamic, thrust, and recovery loads. This subsystem provides the mounting interfaces necessary to support all other subsystems and ensures mechanical and dimensional compatibility throughout the vehicle. Additionally, the Structures Subteam must balance strength, stiffness, and weight considerations while adhering to competition and safety requirements to achieve optimal performance.

1.2.3 Project Requirements

For this project, the Structures subteam is focused on advancing the use and testing of 3D-printed materials in aerodynamic components of the rocket, mainly couplers. The goal is to evaluate how additive manufacturing can be applied to load-bearing and aerodynamic structures while maintaining compliance with IREC's flight and safety requirements. This includes developing and testing 3D-printed fins, fairings, and internal supports to assess their strength, dimensional accuracy, and performance under launch conditions. Quantitatively, these components must withstand aerodynamic pressures at speeds approaching Mach 1 and structural loads exceeding several hundred Newtons without failure or deformation. To meet these goals, the subteam must integrate material testing, finite element analysis, and iterative prototyping while ensuring that all manufactured parts fit within the rocket's size and weight constraints defined by IREC. The results of this work will help determine the feasibility of 3D-printed components for future high-stress aerospace applications and contribute to a lighter, more efficient airframe design for SCRT's rocket.

1.2.4 Table of IREC Rule Restrictions

For the rule restrictions that affect the project see table 7.

Table 7: IREC Design, Test, & Evaluation Guide Restrictions [1]

IREC Section	Requirement	Impact
8.1	Adequate Venting	Must be properly vented to prevent internal pressure build up
8.2	Overall Structural Integrity	Airframe must withstand all stresses during handling launch and flight
8.3	Material Prohibitions	Must use lightweight materials suitable for flight
8.5	Joints Implementing Coupling Tubes	Coupler Joints must extend at least one body-tube diameter into adjoining section

1.2.5 Interaction With Other Subteams

The manufacturing process testing conducted in this project is primarily focused on the Structures Subteam, as its objective is to validate the feasibility of using additive manufacturing for couplers and nosecone as seen in Fig. 3. However, the results of this testing are influenced by other subsystems, particularly sizing, since the forces experienced by the components under evaluation are dependent on the design decisions and operational parameters defined by the rest of the team. Effective coordination between subteams is therefore essential to ensure that the structural components are designed, fabricated, and tested under representative loading conditions.

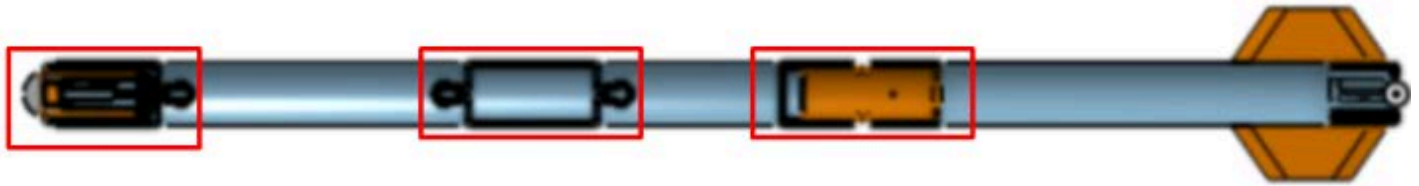


Figure 3: Bounding Boxes Around Parts Affected by Project

2. Current State Analysis and Benchmarking

2.1 Current State Analysis

At present, the team manufactures structural components such as couplers using traditional composite fabrication methods. Because the custom body tube donated to the team features a nonstandard inner diameter, no COTS couplers were suitable without significant modification. Consequently, a student-researched-and-developed (SRAD) coupler was designed and fabricated using woven carbon fiber (Toray T830) with a tailored layup schedule. The separation coupler utilized an 11-layer layup with the sequence 0/45/0/45/0/45/0/45/0/45/0 to accommodate the high axial and shear loads expected during flight, while the nosecone coupler employed a shorter, reduced layup of 0/45/0/45/0/45/0 due to lower loading requirements. Both couplers were manufactured by laying pre-cut Toray T830 woven carbon fiber plies inside a section of the body tube to ensure accurate fit and geometry. The layups were subsequently pressed against the inside of the body tube using a balder bag and cured at 160 °F, followed by final epoxy bonding to secure the coupler within the fore body tube [2]. The completed components are shown below in Fig. 4 and Fig. 5.



Figure 4: Currently Manufactured Coupler

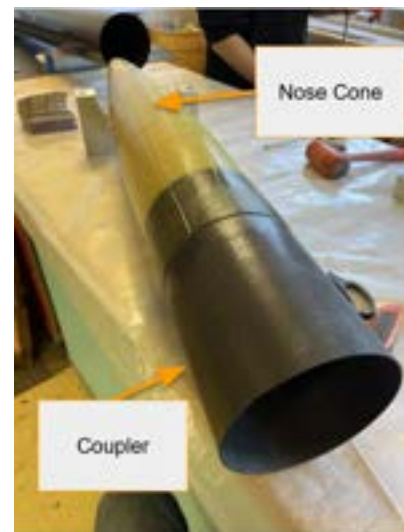


Figure 5: Coupler Nose Cone Connection

This process yields structurally sound components capable of meeting flight load requirements; however, it remains labor-intensive and time-consuming. The reliance on manual layup introduces variability in fiber alignment, resin content, and surface finish, which can lead to inconsistencies between parts. Dimensional deviations also occur as a result of thermal cycling during composite curing. Furthermore, the current approach restricts rapid design iteration and limits geometric complexity, as new molds and layup tools must be physically fabricated for each design revision. Because the team lacks the resources and equipment

to produce complex fixtures, opportunities for innovation and advanced structural optimization are significantly constrained.

2.1.1 SWOT Analysis

For the SWOT analysis of the current manufacturing process see Table 8.

Table 8: SWOT Analysis of Current Manufacturing Process

Strengths	Weakness
• Reliable manufacturing process • Current components have demonstrated adequate strength for competition flight • Long documentation of material properties • Team familiarity with process of production • Current means of accessing required materials for fabrication	• Slower iterative work due to production cycle time • Structural optimization limited by manufacturing constraints • Current fabrication relies on tubing for molding • Issues with thermal cycles affecting dimensions
Opportunities	Threats
• Integration of additive manufacturing to enable lighter optimized components • Faster prototyping for design iteration cycles • Opportunity to blend more additive manufacturing and traditional manufacturing structures • Access to new aerospace additive manufacturing materials	• Limited testing history for SCRT structures • Launch conditions may exceed limits for some additive manufacturing materials • Dependency on printer availability may create production bottlenecks • Budget constraints may affect quality of material that is purchasable • Layer line adhesion when using Additive manufacturing

2.1.2 Current State & SWOT Reflection

This process yields structurally reliable components capable of meeting flight load requirements; however, it remains labor-intensive and time-consuming. The manual layup approach introduces variability in fiber orientation, resin content, and surface finish, resulting in potential inconsistencies between parts. Dimensional deviations may also arise from thermal cycling during the composite curing process. Additionally, the current method limits rapid design iteration and geometric complexity, as each new configuration requires custom molds and layup tooling. These constraints, coupled with limited access to advanced manufacturing resources, restrict opportunities for innovation and prevent the team from fully optimizing structural performance.

2.2 Benchmarking

In previous years, the fabrication of couplers and nosecones relied on a labor-intensive manual layup process using pre-cut Toray T30 woven carbon fiber plies within sections of the body tube to maintain proper geometry. Following the layup, components were cured with epoxy to achieve the necessary structural integrity. While this approach yields strong and reliable parts, the thermal cycles inherent to the curing process often introduce dimensional inconsistencies. Additionally, each coupler demands approximately twelve hours of uninterrupted labor, significantly limiting the team's ability to perform iterative design improvements. Consequently, opportunities to refine component geometry, optimize weight, and explore alternative designs through testing are constrained, making rapid innovation and performance enhancement challenging under the current manufacturing process.

2.2.1 Areas of Possible Improvement

The goal of this Capstone project is to evaluate the feasibility of using additive manufacturing techniques and materials to produce these critical structural components. Additive manufacturing offers the potential to accelerate design iteration, optimize component weight, and reduce fabrication time, while maintaining or enhancing structural performance. Adopting these techniques could streamline the manufacturing process and unlock new opportunities for design innovation, improved subsystem integration, and overall performance enhancement within the rocket's structure. Furthermore, validating a reliable additive manufacturing method and material would provide valuable knowledge for other subteams, enabling them to design complementary components and leverage the advantages of additive manufacturing in their own systems.

2.2.2 Summary Table of Important Benchmarks

For important benchmarks, see table 9.

Table 9: Benchmarking of Current Manufacturing System

Parameter	Current State	Possible Improvement Notes
Material	Toray T30 woven carbon fiber	Access to newer additive manufacturing materials
Manufacturing Time	12 hours hands-on cycle time	Reducing hand-on work
Manufacturing Process	Pre-cut plies laid inside body tube with vacuum bag	Remove large prep time, and need for quality mandrels/molds
Iterative Ability	Due to material cost & labor time, once fabricated, limited ability to iterate	Additive manufacturing allows for iterative design
Innovation/Optimization	No access to molds/fixtures for complex/optimized designs	Additive manufacturing opens up innovation to improve rocket

3. Testing Design Analysis

To achieve a competitive advantage in using additive manufacturing over our previous method of fabrication, a comprehensive understanding of the mechanical properties of the filament materials is required. Given the club's limited resources and the suboptimal quality of the available printing systems, it is essential to evaluate the performance characteristics of the materials that can currently be produced in-house. The objective of this capstone project is to develop and implement a standardized testing methodology for characterizing the mechanical behavior of the filaments in use. The resulting material property data will serve as a reference for design optimization, enabling improvements to critical rocket components and contributing to enhanced overall performance in competition.

3.1 Testing Methods

The first step is to identify the key material properties that should be measured during filament characterization. These properties directly influence how the material can be applied in design and inform how we optimize components that incorporate it. The following sections outline the primary testing methods and measurement targets selected to obtain meaningful data for evaluating material performance and manufacturing accuracy.

3.1.1 Tensile Testing

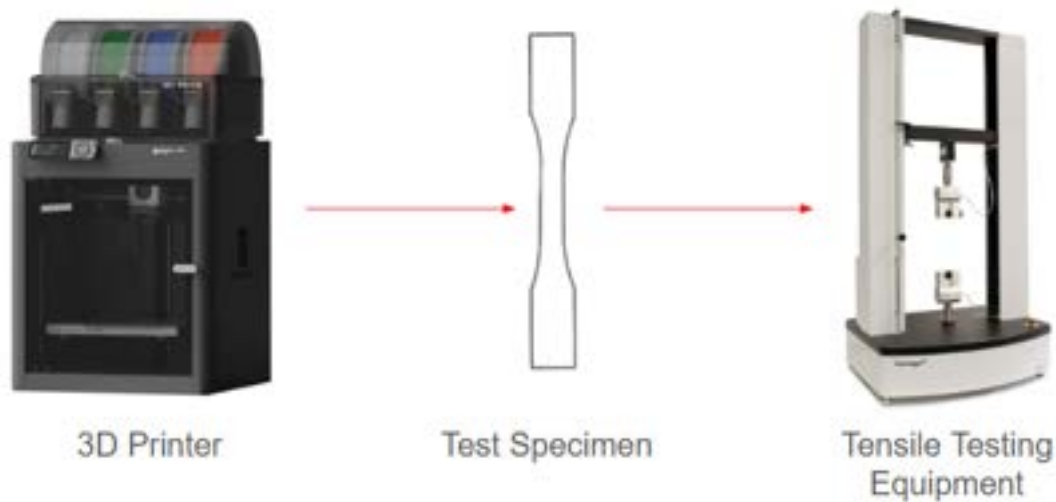


Figure 6: Target 1 - Tensile Testing

One of the primary objectives of this characterization effort is to understand the mechanical behavior of the material under tensile loading. Tensile testing using standardized dogbone specimens will provide key properties such as ultimate tensile strength, yield strength, modulus of elasticity, and ductility. These parameters offer critical insight into the material's performance and limitations, enabling more informed design optimization and ensuring that structural components are engineered with appropriate strength and stiffness considerations. See Fig. 6 above for the basic flow process. Table 10 below analyses the advantages, disadvantages, risks, and failure modes of this testing.

Table 10: Advantage, Disadvantage, Risks, and Failure Modes of Tensile Testing

Advantage	Disadvantage
- Obtaining the yield strength properties allows for data to be used in FEA analysis and design optimization - Helps assess interlayer adhesion and print quality - Should have access to equipment on campus	- Requires producing many test specimens for testing - Needs to have test equipment/machine in order to get accurate data - Does not provide data on fatigue or impact - Time consuming set up - Destructive testing
Risks	Failure Modes
- Improper testing setup - Inconsistent print quality can lead to misleading results - Environmental sensitivity in print can affect the mechanical properties - Limited funding could provide data with lots of uncertainty	- Incorrect Measurements - Operator error - Calibration Errors during setup - Printer inconsistency causing an internal failure - Specimen warping or residual printing stress from production

3.1.2 Dimensionality Measurement

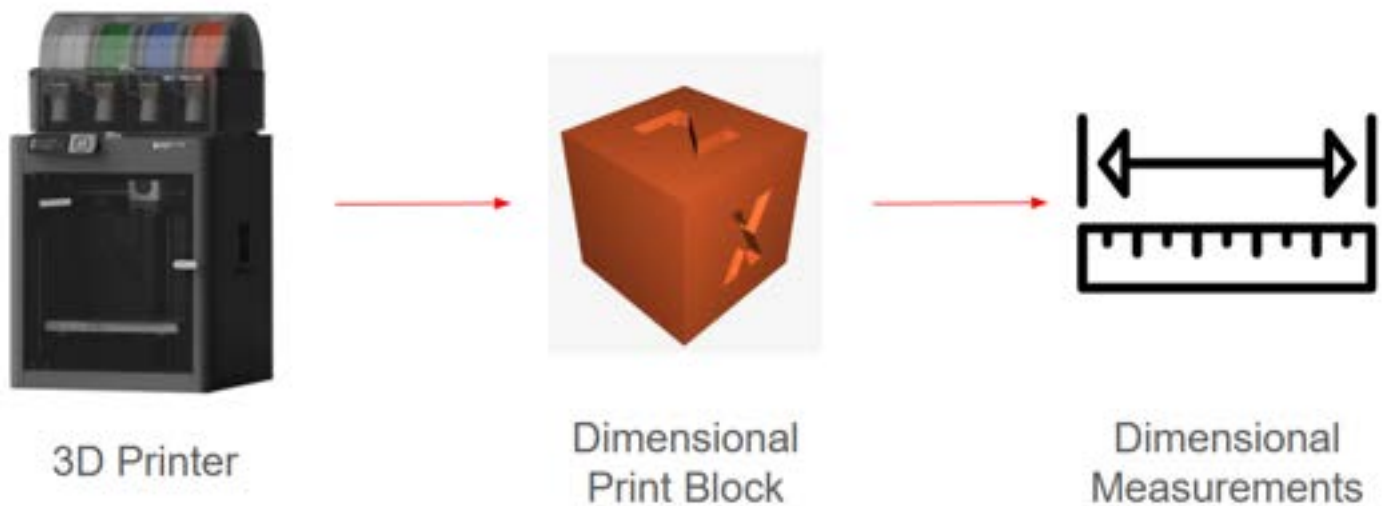


Figure 7: Target 2 - Dimensional Testing

The objective of the dimensional testing is to obtain precise measurements that quantify tolerances and dimensional deviations between the printed part and the original CAD model. This data will be used to determine shrinkage factors and identify any systematic dimensional offsets inherent to the printing process. Understanding these variations enables more accurate modeling, improved print calibration, and better prediction of final part dimensions in future designs. See Fig. 7 above for the basic flow process. Table 11 below analyses the advantages, disadvantages, risks, and failure modes of this testing.

Table 11: Advantage, Disadvantage, Risks, and Failure Modes of Dimensional Testing

Advantage	Disadvantage
• Provides quantitative dimensional data • Enables comparison of print parameters • Enables shrinkage factors and tolerances for accurate design compensation • Non-Destructive Testing • Easily accessible testing methods	• Testing requires much specimen preparation • Testing resolution affects accuracy • Manual measurements will lead to human error • Limited by sample size
Risks	Failure Modes
• Improper testing setup • Inconsistent print quality can lead to misleading results • Environmental sensitivity in print can affect the mechanical properties • Calibration errors	• Incorrect measurements • Tools not calibrated • Data entry errors

3.1.3 Impact Testing

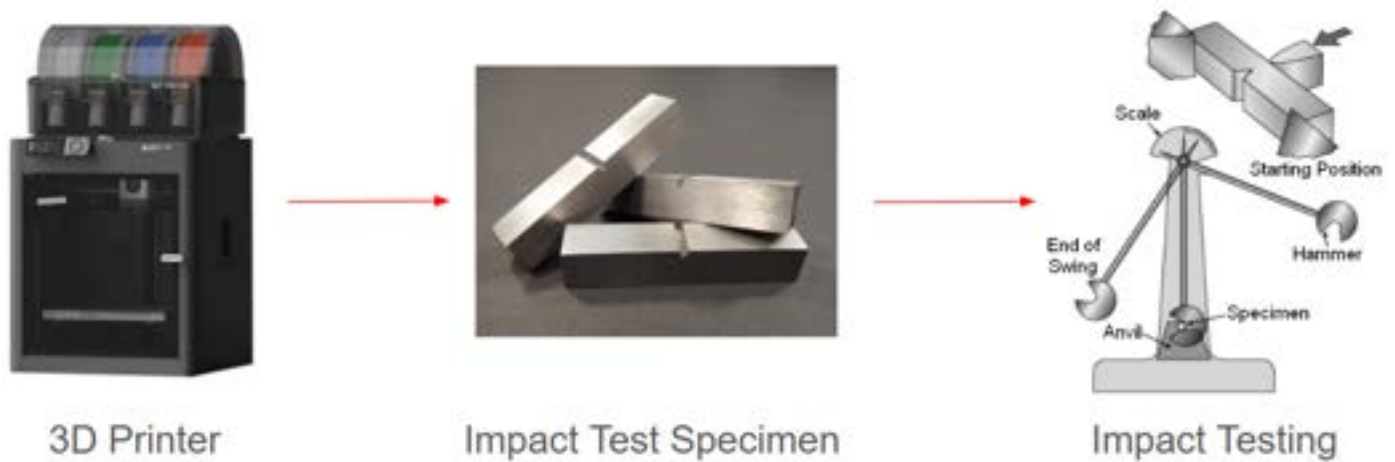


Figure 8: Target 3 - Impact Testing

Impact testing is conducted to measure the material's ability to absorb energy under sudden loading conditions, providing insight into its overall toughness and resistance to fracture. Using a standardized notched specimen and a pendulum-style impact tester (such as the Charpy or Izod method), the absorbed energy during fracture can be measured and normalized by the specimen's cross-sectional area. This value reflects how the material behaves under dynamic stress, which is especially important for evaluating layer adhesion and anisotropy in 3D-printed parts. See Fig. 8 above for the basic flow process. Table 12 below analyses the advantages, disadvantages, risks, and failure modes of this testing.

Table 12: Advantage, Disadvantage, Risks, and Failure Modes of Impact Testing

Advantage	Disadvantage
- Insight into toughness and energy absorption capacity of material - Reveal layer adhesion quality and anisotropy of print orientation - Data easy to compare between different materials	- Testing requires much specimen preparation - Destructive Testing - Requires specialized equipment that is harder to procure - Safety Concerns
Risks	Failure Modes
- Improper testing setup - Machine/Tool accuracy - Inconsistent print quality can lead to misleading results - Environmental sensitivity in print can affect the mechanical properties	- Improper specimen preparation - Human setup error - Instrument miscalibration

3.2 Experimental Variables

Multiple printing strategies can be employed to produce varying material properties and dimensional outcomes. By evaluating a selection of these strategies, we can identify the optimal printing parameters and post-processing methods that maximize mechanical performance, dimensional accuracy, and overall part quality. This knowledge will directly inform design decisions and help ensure that future components are manufactured with reliability and efficiency.

One of the more challenging aspects of 3D printing is the large number of variables that can significantly affect material properties. For our testing, there are several primary variables that are known to have a major impact on print quality and material performance, as well as numerous secondary variables, such as small adjustments in print settings, that can subtly influence the results.

3.2.1 Materials



Figure 9: Example of Different Filaments that Could be Tested

One of the most significant factors influencing test outcomes is the selection of filament material. Each filament type exhibits unique mechanical and thermal properties, which can lead to variations in tensile strength, dimensional stability, and impact resistance. Some examples of filaments can be seen above in Fig. 9. Evaluating multiple filament types will provide insight into their performance under practical printing conditions and help identify the most suitable materials for specific aerospace applications. Additionally, variability in print quality between filaments will be assessed, as certain materials are more consistent and easier to process than others. As seen in Fig. 10, a general process for testing material would look like the sequence outlined below.

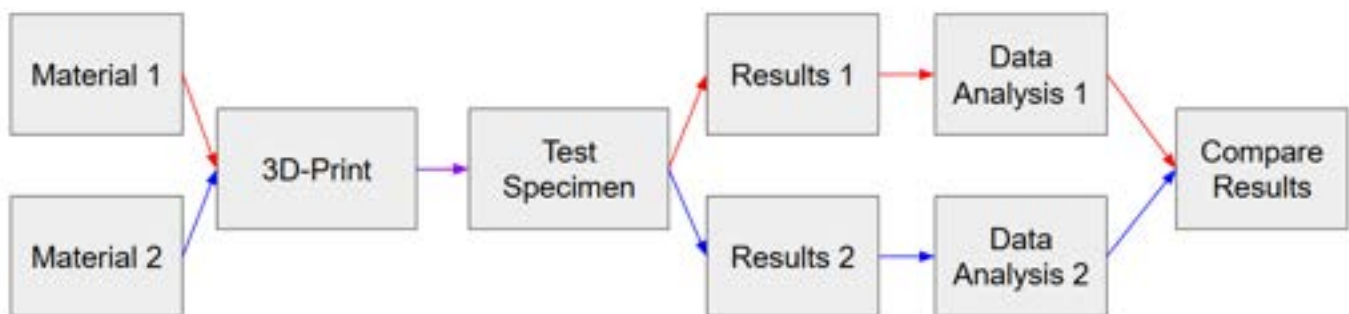


Figure 10: Flow Diagram of Material Testing

3.2.2 Printers



Bambu P1S [3]



Bambu H2D [4]

Figure 11: Examples of Different Printer

The choice of printer can also significantly influence print quality and material performance. See Fig. 11 for the main printers available to us. Differences in hardware, calibration, motion control, and extrusion systems can lead to variations in dimensional accuracy, surface finish, and mechanical properties. By evaluating prints from multiple printers, we can assess how machine-specific characteristics affect results and select the printer that best meets the requirements for a given material or part design. Since the club does not have access to high-end printers, developing a clear understanding of the capabilities and limitations of our available machines is essential to improving print consistency and overall product quality. As seen in Fig. 12, a general process for testing different printers would look like the sequence outlined below.

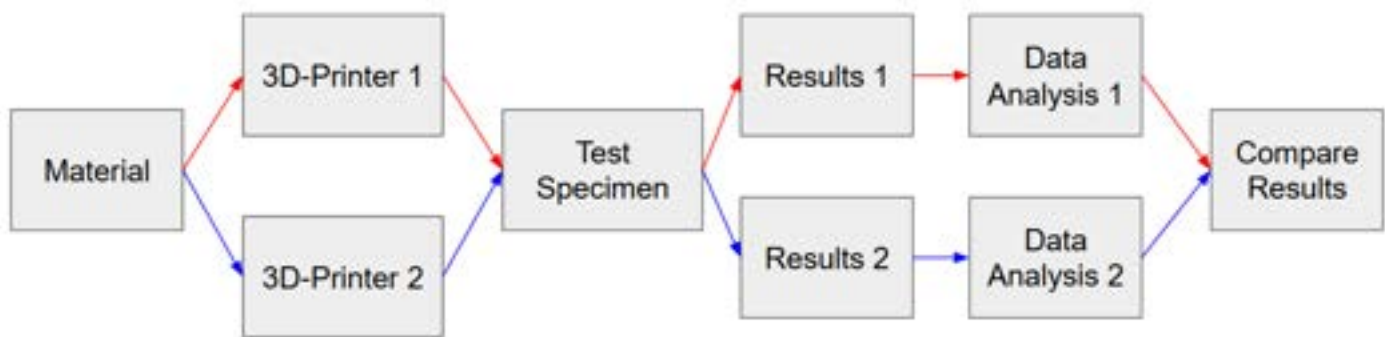


Figure 12: Flow Diagram of Printer Testing

3.2.3 Annealing



Temperature Controlled Kitchen Oven



Annealing Oven

Figure 13: Example of Possible Tool for Annealing

Annealing is a post-processing method that can significantly influence the mechanical and dimensional properties of 3D-printed parts. By heating the material to a controlled temperature below its melting point, internal stresses are relieved and polymer chains have the opportunity to realign, improving strength, stiffness, and thermal resistance. However, annealing can also cause dimensional changes or warping if not carefully controlled. Understanding how different annealing conditions affect our printed materials allows us to refine post-processing procedures and produce more consistent, reliable components [5].

There are several advantages to conducting annealing tests this year. The process is highly accessible for the team, as it only requires access to an oven. Ideally, a unit with precise PID temperature control would be used, since standard kitchen ovens typically exhibit significant temperature variation. Despite this limitation, the low cost and ease of implementation make annealing an attractive and practical option for enhancing both the mechanical performance and dimensional consistency of printed components. As seen in Fig. 14, a general process for testing different annealing processes would look like the sequence outlined below.

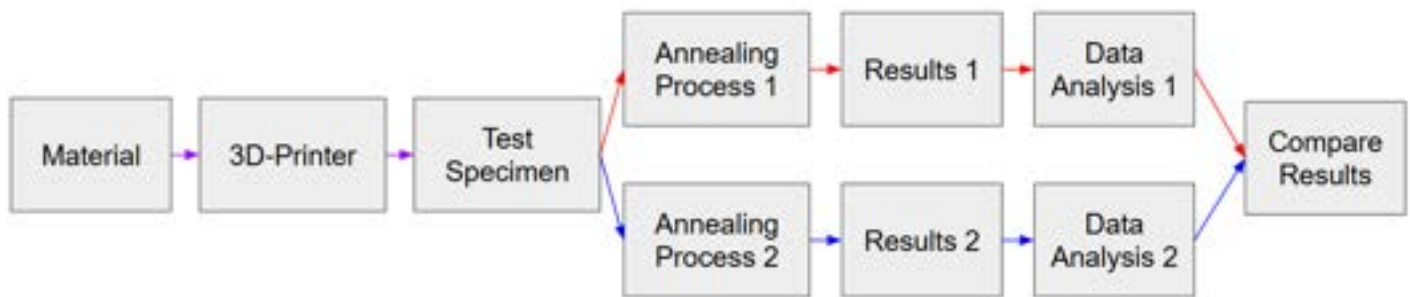


Figure 14: Flow Diagram of Annealing Testing

3.2.4 Print Orientation

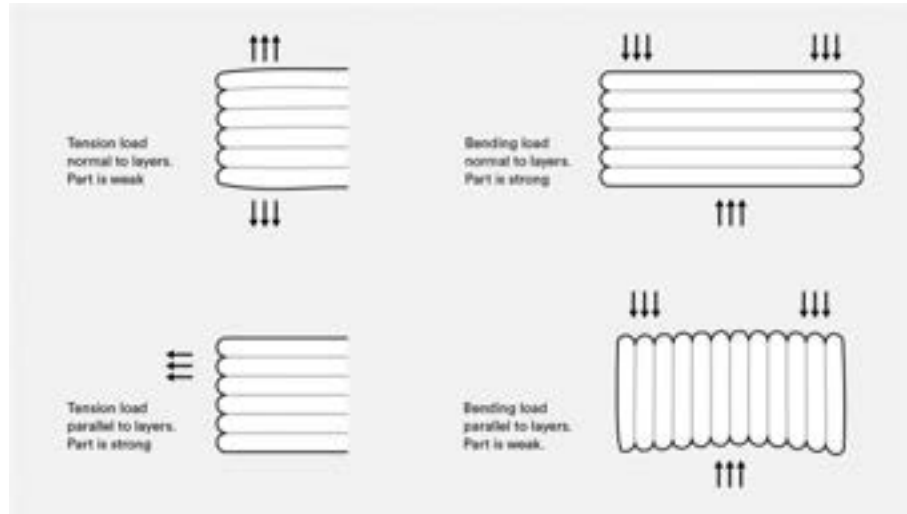


Figure 15: Diagram about print orientation effects [6]

Print orientation plays a critical role in determining the structural and dimensional characteristics of 3D-printed parts. The direction in which layers are deposited influences interlayer adhesion, surface quality, and mechanical anisotropy, often resulting in significant variations in tensile strength and overall durability (see Fig. 15). By testing specimens printed in different orientations, the effects of layer direction on mechanical performance can be quantified, allowing the identification of an optimal orientation for achieving consistent strength, precision, and reliability in future designs. As seen in Fig. 16, a general process for testing printing orientations would look like the sequence outlined below.

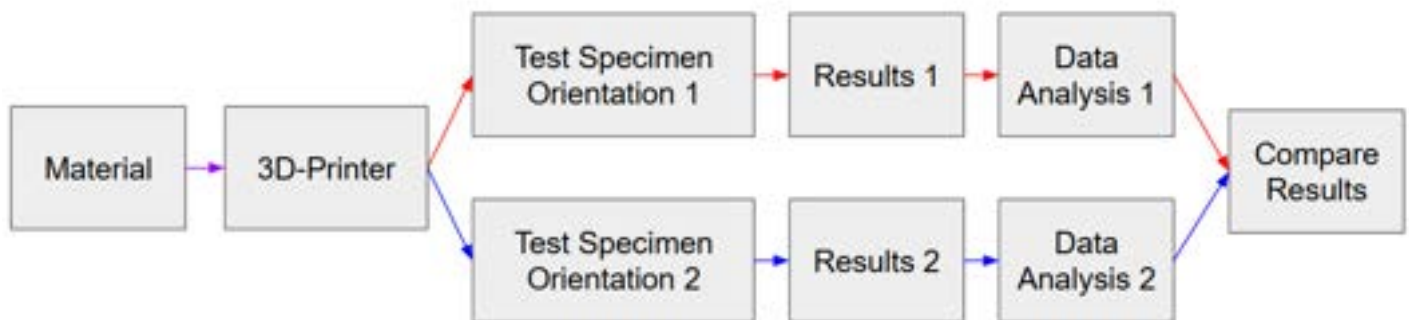


Figure 16: Flow Diagram of Print Orientation

3.2.5 List of Secondary Variables

These are variables known to influence the outcomes of our prints. While a comprehensive understanding of each would be valuable, testing all of them would require an extensive program beyond the scope of our club's objectives. Instead, we focus on the key variables believed to have the most significant impact, while remaining aware of secondary factors that present opportunities for future research and optimization of our designs.

- Print Speed
- Layer Height

- Infill Density
- Nozzle Temperature
- Nozzle Size
- Nozzle Material
- Bed Temperature
- Printer Encasement
- Environmental Conditions
- Filament Batch
- Cooling/Fan Settings
- Filament Color

4. Testing Design Selected

While it would be ideal to conduct all of the proposed testing, limited funding makes this approach unrealistic. This capstone project is one that is not required for our overall rocket to take flight and compete this year, so we are mostly working with the limited available budget. As a result, the project must take a selective and strategic approach in prioritizing testing methods for this first year of material data collection, focusing on those that will have the greatest impact on the team's long-term mission and design objectives.

The primary focus for this year's testing is to determine the tensile strength of the printed materials and assess the dimensional accuracy of the manufactured components. Accordingly, two tests will be conducted in parallel: tensile strength testing and dimensional measurement analysis. These evaluations will provide essential baseline data to assess material performance and manufacturing reliability for future design applications. See Fig. 17 for a summary of the chosen testing process.

For future years, the material testing results will allow the club to perform more accurate FEA on the parts we design. While published data provides idealized mechanical properties, it's critical to understand the real characteristics of the components we can actually produce. With this knowledge, we can design higher-quality, lighter parts that improve performance and reliability, ultimately making the team more competitive in future IREC events.



Dimensional Measurement



Instron Tensile Test



3D-Printer



Annealing Oven Time

Figure 17: Important Testing and Changing Variables

4.1 Rationale for Testing Selection



Figure 18: Integrated plan for Printer and Annealing Variance

For this year, based on the resources available to our team, the number of tests that we can do are very limited. One of the shortcomings of most material testing processes is the fact that it is at its core destructive testing to find when material fails. For that reason, it requires several test specimens and in turn higher cost to procure enough filament to get data that we can be confident in. Thus, even though we have two tests that would provide a lot of data in the tensile and the impact testing, we are going to only do tensile testing as the destructive test for this year. This was agreed upon by the team lead in the club, as the values obtained from tensile testing are critical for the FEA analysis and component optimization that the club wants to work towards the next year.

As seen above in Fig. 18, the flow diagram shows the general process that we are going to be targeting for this year. With dimensional testing, by using the same specimens for tensile testing, we are going to do non-destructive dimensional analysis in order to quantify the shrink factors that we are seeing produced by our processes. Then taking those same specimens, we will run through tensile testing in order to get its modulus of elasticity as well as its ultimate yield strength. These values are the ones that are the most critical to producing high quality and optimized parts, and given the limited funding is the most achievable with our given resources.

For our tests, we set out to evaluate two different 3D printers, the Bambu P1S and the Bambu H2D, in order to compare the quality of parts they produce. The primary distinction between the two systems lies in their enclosure design and the resulting temperature conditions during printing. Differences in chamber temperature, airflow, and internal hardware can influence layer adhesion, dimensional stability, and surface finish. By comparing print quality and dimensional accuracy between the two machines, we aimed to determine which printer produced more consistent results and establish the shrinkage factors necessary for achieving tighter tolerances. However, due to circumstances that prevented the use of the H2D, the project ultimately proceeded using only the available P1S.

After printing the specimens on the printer, a portion will be retained as control samples while the remaining specimens will undergo annealing. The annealing process will be conducted at fixed durations of 0, 3, 6, 9, and 12 hours to evaluate how heating time influences both material properties and dimensional stability. By comparing results across these intervals, we aim to determine the point at which improvements in mechanical performance and dimensional accuracy converge. This will allow us to establish an optimal annealing duration and avoid excessive processing that provides negligible benefit.

4.2 Technical Specification

There are several parameters that must be standardized to ensure consistency across all tests. Among the most critical are the specimen geometry and the general print settings. These serve as baseline control variables, allowing us to minimize variability in the printed samples and ensure that any observed differences in performance are due to the factors under investigation rather than inconsistencies in fabrication.

4.2.1 Benchmark for PA6-GF

Below is the Benchmarking of PA6-GF as seen in table 13.

Table 13: Benchmarking of PA6-GF

Parameter	Values
Young's Modulus (Z)	1950 +/- 210 MPa
Tensile Strength (Z)	27 +/- 5 MPa
Shrink Factors	None

4.2.2 Testing Specimen

For tensile testing, specimens will be prepared in accordance with ASTM D638. The team will follow the Instron testing guidance and select from the five allowable specimen geometries defined in the standard [7]. Type I specimens will be used, as this geometry is the standard configuration recommended for most thermoplastic tensile evaluations. The shape and dimensions of the Type I specimen are shown in Fig. 19.

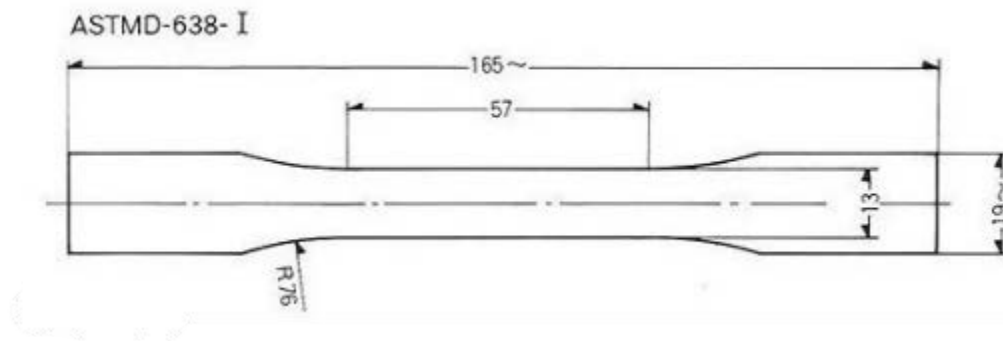


Figure 19: ASTM-638-I Specimen Dimensions in Millimeters [8]

For more specific dimensional requirements for the tensile specimens, refer to table 14 below.

Table 14: ASTM D638 Specimen Type 1

Variable	Length
Full Length	165 mm
Parallel Length	57 mm
Gauge Length	50 mm
Parallel Section Width, Strong	13 mm
Thickness	4 mm
Grip Section Width, Strong	19 mm
Distance Between Grips	115 mm

4.2.3 General Print Settings

To ensure consistency across all printed specimens, we will maintain uniform printing parameters wherever possible. Table 15 summarizes the critical print settings selected for standardization [9]. An additional essential step prior to printing is proper filament drying. Because moisture absorption can significantly affect

print quality and mechanical performance, each filament spool will be dried before use. While we do not currently have a formal drying standard, the target process is to store filament in a drying unit for approximately 12 hours before printing.

Table 15: Settings to be Set Consistent Across Printers

Variable	Value
Printing Speed	110 mm/s
Nozzle Temp	290 °C
Layer Line Height	0.2 mm
Build Plate Temp	110 °C
Part Cooling Fan	20%
Part Print Orientation	Z-Axis

4.2.4 Annealing Standard

We will also use a consistent annealing procedure for all specimens. Each part will be processed in an annealing oven maintained at 110 °C to ensure uniform thermal exposure and repeatable post-processing conditions.

4.3 Documentation

To support future analysis and continued development of the club's manufacturing capabilities, all data collected in this project must be clearly and consistently documented. This section defines the standards for recording print settings, material batches, specimen preparation, and test results.

4.3.1 Specimen Creation Tutorial

The fabrication of standardized specimens is the first step in the material characterization process. The following procedure outlines the required steps to prepare, print, and document each specimen to ensure consistency and repeatability across all samples.

1. Purchase Material from Vendor such as Bambu Lab
2. Dry Filament for minimum 12 hours 80 degrees celcius in drying container at or enclosure before use
3. Load Filament Into Printer
4. Take CAD model of ASTM D638 Type I Testing Specimen and Implement into Slicer in the Z-axis direction
5. Apply Standardized print setting as listed in table 15
6. Export G-Code from Slicer to upload to select printer
7. Print specimen on printer (see Fig. 20 for completed specimen)
8. Set aside some specimen as control specimen
9. Label Each specimen with its printer type, as well as planned annealing duration



Figure 20: Completed Specimen

4.3.2 Annealing Tutorial

After fabrication, the specimens undergo a controlled annealing process to evaluate how heat treatment influences mechanical performance and dimensional stability. The following procedure outlines the steps used to anneal each sample under consistent conditions.

1. Preheat annealing oven to 110 °C
2. Set annealing time to desired annealing duration
3. Place specimen in temperature-stable location in oven
4. Remove specimen from oven when time set is finished
5. Let specimen cool and re-label specimen with printer and annealing duration

4.3.3 Dimensional Testing Tutorial

To evaluate how printing and annealing affect part geometry, each specimen was measured for key dimensions before mechanical testing. The following instructions summarize the dimensional inspection procedure.

1. Measure Each specimen in the X, Y, and Z direction using calipers
2. Compare measured value to control specimen
3. To find shrink factor, take measured value over control value.

4.3.4 Tensile Testing Tutorial

To characterize the mechanical performance of the printed and annealed specimens, tensile tests were conducted in accordance with ASTM D638. The steps below describe testing configuration, instrumentation, and procedures used to obtain repeatable and comparable results.

1. Mount specimen into tensile tester
2. Run Instron machine test
3. Record stress strain data

4. Extract modulus of elasticity, and yield strength
5. Log all results into data sheet

5. Implementation

To implement the planned testing, the sections below cover the manufacturing process followed, as well as issues encountered.

5.1 Manufacturing Process

5.1.1 3D CAD of Test Specimen

Using standards from ASTM D638, the CAD model seen in Fig. 21 was created.

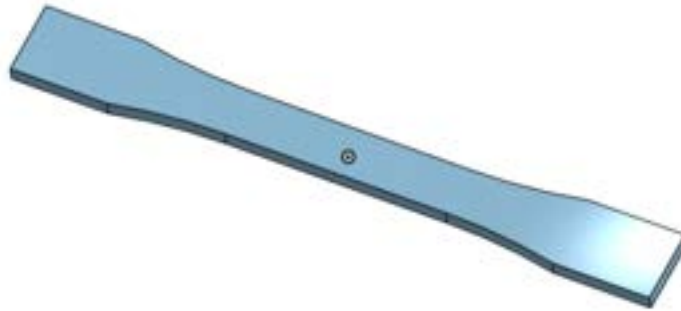


Figure 21: CAD Model of Test Specimen

5.1.2 Printing of Specimen

Using the 3D printers available to the club on Oregon State's Campus, Test Specimens were printed with the Z-axis being the testing axis as it is historically the weakest direction. Below in Fig. 22 the specimen in the slicer is shown.

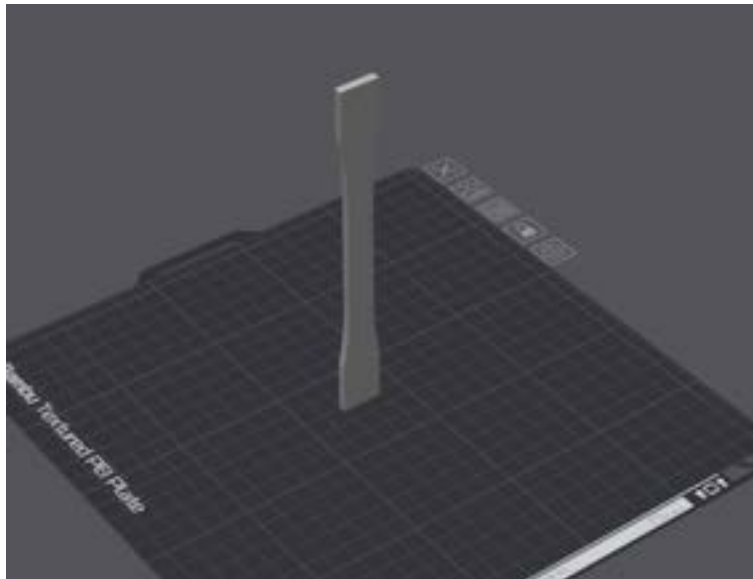


Figure 22: Printing Orientation of Specimen

The results of the prints can be seen in Fig. 23 below.



Figure 23: Printed Specimen

5.1.3 Initial Dimensional Testing

To get baseline data, the first step is to label all of the specimens and then take their initial dimensional measurements. This was done with a caliper, and all measurements were listed down into an excel file. These values will be used for shrink factor calculations and be used to compare dimensional sizing changes caused by annealing. Below in Fig. 24 is a photo of the caliper used for measurements alongside a specimen.



Figure 24: Caliper used for Dimensional Testing

5.1.4 Annealing Specimen

To perform the annealing process on our specimen, the annealing ovens accessible through Ilabs were used. Setting the temperature to 100 degrees celsius, the specimens were placed inside and left to anneal for different set time periods discussed previously. When the annealing time was reached, the specimens were left

to cool and stored away to be measured for potential shrinkage. Below in Fig. 25 and Fig. 26 are the annealing oven and the setup for annealing



Figure 25: Annealing Oven



Figure 26: Annealing Setup

5.1.4 Tensile Testing Process

To get the material properties using an instron machine we tested each of the sets of specimens. The specimens were minted into the tensile tester and pulled until failure. These values were tracked by the machine giving us an understanding of the stress and strain that then calculates the modulus of elasticity in the Z-axis direction. Below in Fig. 27 is an image of the testing.



Figure 27: Tensile Testing of the Specimen

5.1.5 Analysis of Tests

The qualitative fracture analysis provided additional insight beyond the measured mechanical properties. Across the specimen groups, failure occurred in noticeably different locations and modes. Several samples fractured cleanly at the smallest cross-sectional area, aligning with the expected stress concentration predicted by the geometry. Others, however, exhibited multiple fracture sites or irregular break patterns, suggesting the presence of residual stresses introduced during manufacturing or cooling. These variations in

failure behavior highlight potential inconsistencies in print quality and internal stress distribution. Representative fracture surfaces are shown in Fig. 28.



Figure 28: Specimens After Testing (Annealing Groups from Left to Right)

5.2 Issues Encountered

During Production of the specimen, many issues were encountered. The main issues faced can be seen in the sections below. There were also logistical issues faced that delayed and prevented timely completion of this capstone project.

5.2.1 Print Adhesion

One of the main issues faced in production was keeping the first layer of filament adhered to the printing surface. Even using a standard textured plate. As the print raised to about 25% of its height, which translates to about 40mm high, the specimen would slightly move causing printing failure. This failure can be seen below in Fig. 29. In order to increase the adhesion to the bed plate, a few methods were attempted. From previous years, white PA6 was noted to be especially difficult to print quality parts. For the official specimen, we choose to move over to an orange colored filament. For more improved adhesion, glue was also used on the textured plate.

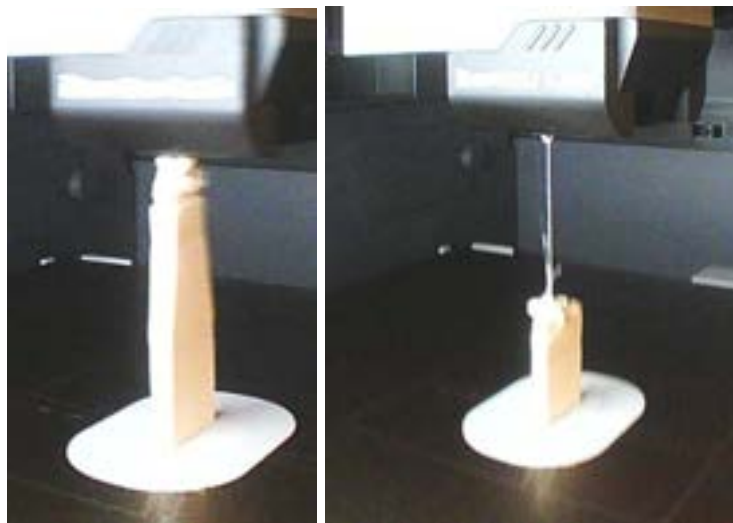


Figure 29: Print Failures Caused by Bed Adhesion

5.2.2 Filament Moisture

Another issue that was observed was low quality prints due to the filament not being dried out enough. In order to get high quality prints, the filament must be dried to remove excess moisture. For the first prints, the dryer we were using could not get to the acceptable temperature to get the moisture percentage low enough resulting in prints that had poor layer adhesion and flaky parts. The low quality prints can be observed in Fig. 30 below.



Figure 30: Low Quality Print Attributed to Moisture in Filament.

5.2.3 Filament Cooling

Similar to the issues above with the filament moisture content. Another issue encountered was insufficient cooling of the filament causing issues with the layer lines. This error would cause unevenness in the print that was then caught by the printer head causing delamination from the print bed. In Fig. 31 below, the print failure seen on the far left was attributed to insufficient cooling. In order to combat this issue, in the slicer, the fan was increased to a higher rpm in order to have more cooling.



Figure 31: Delaminated Print Caused by Insufficient Cooling

5.2.4 ILabs Facility Access

To complete the 9 and 12 hour annealing times, the specimen would need to be left in the annealing oven past the 5pm closing of Oregon State Universities ILabs facilities. Getting this access got majorly delayed, and prevented me from working on my project as I awaited after hours approval. Once approval was completed, the rest of the capstone project went smoothly

6. Data Results

6.1 Methods of Data Collection

The two methods of data collection are using calipers for dimensional measurements, and then instron tensile tester for the material properties. In order to increase the chances or better accuracy each of the tools were properly calibrated. For dimensional measurements, each piece was measured more than once to check for consistency. For the instron, multiple specimens were tested in order to get more accurate data and to check for outlier cases.

6.2 Tests Complete to Date

6.2.1 Dimensional Testing

As mentioned in section 5.1.3 the use of a calibrated caliper was used to get the dimensions in the x, y, and z axis direction. These dimensions were measured at every stage, from initial print to post annealing in order to get an understanding of dimensional changes throughout the manufacturing process. This allows for predicted shrink factors that can be used in the slicer to get the correct final dimensions.

6.2.1 Tensile Testing

For tensile testing, the instron machines that are accessible though ILabs were requested to use. This machine performs a tensile test by pulling apart test specimens until failure, while keeping track of the stress being applied, as well as the strain of the material. Those values then give the modulus of elasticity an important material property for future development of our project.

6.3 Results of Testing

Following the completion of testing, all dimensional and tensile-strength data files were archived in the team's drive for use in future years. The final averaged values derived from this dataset are summarized in table 16. Full data results have been stored in the teams google drive.

Table 16: Summary of Key Results

Annealing Time (hrs)	Shrink Factors	Tensile Strength (MPa)	Young's Modulus (MPa)
0	0.9929	48.75	1655.15
3	0.9926	47.12	1619.51
6	0.9922	48.01	1551.23
9	0.9917	43.33	1608.63
12	0.9913	42.37	1712.06

Compared to the benchmarking values provided in the filament datasheets (Table 13), the printed specimens exceeded the expected tensile strength but showed a slight underperformance in Young's modulus. This indicates that while the material was able to withstand higher ultimate loads than predicted, its stiffness remained marginally below the manufacturer's specification.

For shrinkage behavior, the testing demonstrated that the material exhibits minimal dimensional change. Even after annealing, the specimens experienced only slight reductions in size, confirming that shrink factors for this filament are small and largely negligible for most geometries.

7. Conclusion

The completion of this capstone project was carried out independently, and the experience included both successes and challenges. The testing process and the resulting data were major accomplishments, and they formed the core of the project's technical contribution. At the same time, several aspects of the work were made more difficult by broader logistical issues within this year's capstone structure. Navigating these obstacles required flexibility and persistence, and reaching the final set of results remains something I am proud of.

One of the largest challenges involved the lack of clarity surrounding funding and available resources. For several weeks, it was difficult to define the scope of the project because the budget was not clearly communicated. This uncertainty affected decisions about materials, testing plans, and the overall scale of the work. A more transparent understanding of what resources were available would have made it much easier to plan the project and move forward efficiently. Clearer communication in this area would also help future students avoid delays and begin their work with a more realistic sense of what can be accomplished within the given constraints.

Another challenge was the absence of consistent guidance during the early stages of the project. Without a project advisor to help evaluate the proposed scope, it was difficult to determine whether the planned work was appropriately sized for the available timeframe. This lack of early feedback created uncertainty about how ambitious the project should be and made it harder to balance depth with feasibility. Having additional perspectives at the beginning would have helped refine the direction of the project and prevented unnecessary setbacks.

Despite these difficulties, the experience provided valuable insight into the realities of engineering work. Managing limited resources, adapting to unexpected constraints, and maintaining progress without continuous oversight are all skills that translate directly to professional environments. Completing the project under these conditions reinforced the importance of clear communication, proactive planning, and steady problem solving.

7.1 Use Case

The measured shrinkage values create a reliable foundation for improving dimensional accuracy in future PA6-GF prints. These values allow designers to compensate for predictable changes in size during cooling, which can reduce the amount of post-processing and rework required. As the dataset grows in future years, these adjustments can become even more precise, especially for parts with complex geometries or tight tolerances.

For smaller components that can fit within the available annealing ovens, the results also support the idea of integrating annealing into the manufacturing workflow. The dimensional changes observed during annealing were small and consistent, which means the process can be applied without introducing uncertainty into part fit or assembly. Annealing may also help reduce internal stresses that develop during printing, which can improve long-term stability and reduce the likelihood of warping during service.

The tensile strength and modulus data collected during testing provide additional benefits for design and simulation. By using these experimentally measured values, future teams can create custom material

datasets for FEA that more accurately represent the behavior of printed and annealed parts. This approach avoids relying on idealized datasheet values that may not reflect the actual performance of the material when processed through additive manufacturing. The measured properties were generally lower than the manufacturer's specifications, so using these conservative values in simulation helps reduce the risk of unexpected failure. Designs that pass analysis with these realistic inputs are more likely to perform reliably in real-world conditions.

These findings also influence decisions about part geometry, safety factors, and load paths. With a clearer understanding of how PA6-GF behaves under tension, designers can make more informed choices about wall thickness, infill patterns, and reinforcement features. Over time, this can lead to a library of validated design practices that improve the consistency and reliability of printed components.

The combination of shrinkage data, mechanical testing, and improved simulation accuracy supports the development of a more standardized manufacturing process. As future teams expand the dataset and refine the workflow, the group can move toward a repeatable and well-documented method for producing structural parts that meet the performance needs of the project.

7.2 Advice for Future Years

For future years, this capstone project has considerable potential for growth and continued refinement. The limited scope of this year's work, which was shaped by several external constraints, meant that many of the planned ideas could not be carried out. These uncompleted tasks provide a clear starting point for later teams. By expanding the testing matrix, collecting larger datasets, and validating more variables, future groups can build a stronger foundation for understanding how PA6-GF behaves under different manufacturing conditions. This additional information can be used to optimize print settings, improve dimensional accuracy, and produce structural parts with more predictable performance.

There is also meaningful value in exploring alternative filament types. Comparing PA6-GF with other engineering-grade materials would allow future teams to evaluate tradeoffs in strength, stiffness, shrinkage, printability, and cost. A broader comparison across multiple materials would help determine which filament is best suited for specific applications within the team's projects. This type of work would also support the creation of a more complete material database that future students can rely on when designing parts.

Beyond technical improvements, future teams can benefit greatly from building stronger connections with campus resources. Oregon State iLabs manages many of the machines and testing tools that are essential for collecting accurate and reliable data. Establishing relationships with the staff early in the project can help secure access to equipment, reduce scheduling delays, and open opportunities for additional testing that may not have been considered initially. These connections can also provide guidance on best practices for using specialized equipment, which can improve the quality of the results.

Another important lesson for future teams is the value of setting realistic and well-defined goals at the beginning of the project. One of the challenges I encountered was the absence of a project advisor who could help determine the appropriate scale of the work. Without additional perspectives, it is easy for a project to become either too ambitious or too limited for the available timeframe. Early feedback from an advisor or mentor can help ensure that the project is balanced, achievable, and aligned with the team's capabilities and resources. This type of guidance can also help prevent unnecessary setbacks and keep the project on a steady path.

Finally, future teams may consider documenting their process in greater detail. Clear records of print settings, environmental conditions, testing procedures, and unexpected issues will help create a more reliable reference for the next group. Over time, this documentation can evolve into a comprehensive guide that supports consistent manufacturing practices and reduces the learning curve for new students.

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